

# **Unit 4 :: Data Preparation and Management**

☞ SEC-03-PHY-91-R-03 ☞

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# Syllabus

## Unit 4: Data preparation and management

Basic pre-processing of data: filtering, sorting, normalization and handling missing values; managing multisheet workbooks and structured datasets, importing data from diverse file formats (CSV, ASCII, DAT), and ensuring interoperability between different spreadsheet and analysis platforms.

# 1 Importing data from diverse file formats

Suppose a physics experiment records a variable which is of interest to us, such as temperature, pressure, voltage, count rate etc. These variables are recorded (or sampled) at a sequence of time instants. Now this raw record of data produced by an instrument is almost certainly unfit for direct analysis because

- a) It may contain unreliable readings.
- b) It may be out of chronological order.
- c) It may mix standard and non-standard units.
- d) It may contain gaps.

Then we say that such a dataset must be *pre-processed* before it can be visualised or fitted to a theoretical curve.

For the purpose of pre-processing, the dataset must be first stored and archived in the form of a file. Three file formats are widely physics laboratory exports:

1. Comma-separated values (CSV files)
2. Whitespace-delimited plain text (ASCII files)
3. Instrument-specific data (DAT files)

## 1.1 CSV files

A comma-separated values (CSV) file is a plain text data format for storing tabular data where the fields (values) of a record are separated by a comma and each record is a line (i.e. newline separated). For example, the following are the contents of a CSV file:

```
1 id, name, email
2 1, John, john.doe@example.com
3 2, Jane, janey72@test.org
```

In software that generally deals with tabular data such as a database or a spreadsheet, the contents of the above CSV file is parsed in the form of a table as follows:

| id | name | email                |
|----|------|----------------------|
| 1  | John | john.doe@example.com |
| 2  | Jane | janey72@test.org     |

The first line is used as the table header and the remaining lines are used as the rows of the table.

**1.1.1 Steps to create a CSV file in Windows**

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**1.2 ASCII files****1.3 DAT files**

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